

ERC Consolidator Grant 2023
Research Proposal [Part B1]

ALPHABETS FOR QUANTUM FIELD THEORY

ABCs

PRINCIPAL INVESTIGATOR:

Matthew von Hippel

HOST INSTITUTION:

*Institute of Theoretical Physics,
CEA Paris-Saclay*

DURATION:

60 months

ABSTRACT:

A new era of precision physics demands new tools for evaluating Feynman diagrams, which are important both to collider and gravitational wave physics. The mathematical formalism of symbol alphabets has led to great progress, but these alphabets are currently well understood only for nested integrals of rational functions. Diagrams involving functions beyond these can be described using geometric spaces, from tori in the simplest cases all the way to Calabi-Yaus, and such diagrams are essential for near-term precision physics. To evaluate them efficiently and usefully, I propose to develop and test a new symbol alphabet formalism: one which is mathematically consistent, captures the physical complexity of the Feynman diagrams, and which takes inspiration from recent discoveries in theoretical high-energy physics, particularly in the theoretical laboratory of planar maximally supersymmetric Yang-Mills theory.

A

EXTENDED SYNOPSIS OF THE SCIENTIFIC PROPOSAL

A.1

BACKGROUND AND MOTIVATION

A.1.1

THE CHALLENGE OF PRECISION

Ten years ago, two events opened new realms of precision physics. The 2012 discovery of the Higgs boson was an impressive confirmation of the Standard Model, but was in a sense merely the beginning: today, the Large Hadron Collider has seen only five percent of its planned data. Future measurements will allow us to probe the properties of the Higgs boson, observing whether its interactions and those of the W and Z bosons hide new physics. Later, LIGO's 2015 detection of gravitational waves opened a window on an entirely different area. Future gravitational wave telescopes, such as LISA and the Einstein Telescope, will open the gravitational sky to tests of even greater precision, giving new constraints on the behavior of black holes and the equation of state of neutron stars.

This new emphasis on precision places enormous demands on theory. At the LHC, fully exploiting the new data will require calculation of Feynman diagrams at second and even third order in perturbation theory (in particle physics jargon, two or three loops) [1]. These computations are especially challenging due to the prominent role of particles, such as the top quark, W, and Z, whose masses cannot be neglected in the calculation. Surprisingly, Feynman diagram calculations are also important for gravitational wave astronomy, where techniques from quantum field theory have demonstrated impressive promise [2]. In particular, probing the equation of state of neutron stars will require Feynman diagrams with at least five loops.

In both arenas, the core of the challenge is to compute **scattering amplitudes**, sums of Feynman diagrams used to compute probabilities for particles to scatter off of each other. Often, one needs to compute these sums of diagrams *analytically*: while general numerical methods can suffice for a first pass, an analytic understanding yields both dramatically faster numerics and the ability to reuse results across different situations.

The last decade has seen enormous progress in computing scattering amplitudes analytically at high loop orders. Much of this progress relies on expressing these amplitudes in a particular form, as nested, or **iterated** integrals of rational functions. All one-loop Feynman diagrams can be expressed in this way, as can many higher-loop diagrams involving massless particles. Building off the nested structure of these integrals allows one to decompose them into “words”, or **symbols**, written in an **alphabet** of letters corresponding to the functions' branch cuts [3]. Knowledge of this alphabet has powerful consequences, allowing us to re-express a Feynman diagram with different kinematics, find fast-converging series expansions, and even in some cases “bootstrap” a scattering amplitude from its behavior in kinematic limits without ever needing to draw a Feynman diagram [4]. The concept also served as a foundation for a differential equation method that is currently the gold standard for computing multi-loop Feynman diagrams for collider physics and gravity [5].

It would be fantastic if we could take advantage of this progress indefinitely, but an obstacle stands in our way: iterated integrals *beyond rational functions*. A variety of Feynman diagrams, especially those involving massive particles, *cannot* be represented involving only iterated integrals of rational functions, including many examples at two and three loops directly relevant to near-term collider [6] and gravitational wave physics [7]. These new integrals instead can be associated with geometric spaces, with the more familiar zero-dimensional branch point of a logarithm spread out into a space of multiple dimensions. The simplest such integrals beyond the rational are associated with a torus, while more complicated integrals can involve even higher-dimensional geometries. Surprisingly, many of these geometries turn out to be Calabi-Yau, a type of geometry studied by string theorists as candidate shapes for extra dimensions of space-time.

As an example, consider the Yukawa coupling between the Higgs boson and the top quark, the parameter of the Standard Model which gives the top quark its mass. A precision measurement of this coupling at a hadron collider will involve measuring processes where two partons (for example, gluons) collide, emitting a Higgs boson, a top quark, and a top anti-quark. The two-loop diagram in Figure 1 is one contribution to this process. If one wanted to evaluate this diagram using current methods, one would swiftly encounter an obstruction corresponding to a two-dimensional Calabi-Yau, referred to as a K3 for their loose resemblance to the Kashmiri mountain K2. I and others have characterized this obstruction [8, 9, 10, 11], but as of yet there are no efficient methods to compute diagrams of this kind. To find them, we will need something akin to the symbol alphabet for this new class of functions.

A symbol alphabet has been proposed for iterated integrals involving tori, the first class beyond the rational case [12]. This alphabet formalism has seen some productive early use, identifying it as a natural starting point for generalization. Based on the torus proposal, I developed an alphabet for a particular family of diagrams involving Calabi-Yaus, called **sunrise graphs** [13]. These graphs are corrections to the propagator of a massive particle, and appear very generically as parts of Feynman diagrams. (They are called sunrise diagrams because the particle lines resemble a sun rising over the ocean.) Doing this, I found that the procedure used to find

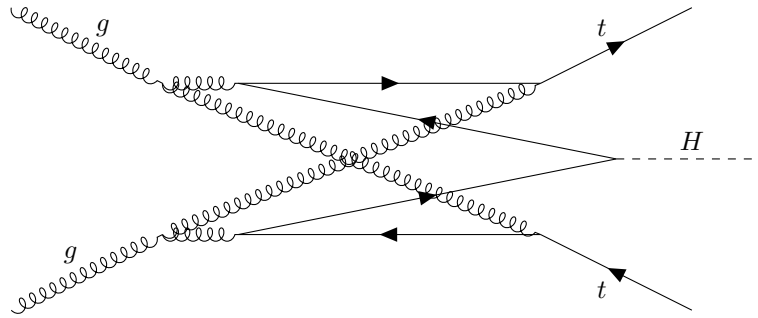


Figure 1: Feynman diagram contributing to a future percent-level measurement of the Higgs-top quark Yukawa coupling, which gives the top quark its mass. To evaluate this diagram, we must understand a geometric space called a K3.

the original torus symbol alphabet, when generalized, was actually ambiguous: by making different choices, one could find radically different alphabets. We will need to make these choices in a *useful* way, to find an alphabet that can address upcoming challenges in precision physics.

A.1.2

THE CLUE IN SUPERSYMMETRIC YANG-MILLS

To find the most useful choices, I will make use of a guide that might seem unusual: a theory with not just supersymmetry (symmetry under exchange of bosons and fermions), but also an infinite number of colors of gluons. This theory, called planar maximally supersymmetric Yang-Mills (MSYM), can be thought of as a cousin of Quantum Chromodynamics, one with dramatically more varieties of particle but that is in other ways much simpler. (For example, the theory is invariant under changes of scale.) MSYM directly describes neither collider physics nor gravitational waves, but developments in it have historically been useful in both, as the calculations required can often be quite similar. It was developments in this theory that first introduced symbol alphabets to the physics community [3], leading soon after to the modern standard for differential equation methods for Feynman diagrams [5]. Subsequently, it was discovered that the symbols of amplitudes in this theory have unexplained structure. Their alphabets obey a mathematical relationship called a **cluster algebra** [14] which constrains which letters can occur next to each other, demanding that they occupy the same “cluster” [15]. Some calculations in the theory even possess a strange duality, which maps them to other calculations or to themselves when the order of the symbol is reversed [16, 17].

These properties should be more than mathematical curiosities: as properties of physical scattering amplitudes, they should have some physical meaning. In order for this to be true, they must have some validity beyond iterated integrals of rational functions, since scattering amplitudes in this theory can involve not just tori, but Calabi-Yaus of arbitrary dimension [18]. However, this will not be the case for every possible symbol alphabet formalism one can propose for these functions. Instead, the choice of alphabet formalisms for integrals beyond the rational will be constrained by the properties of MSYM in a particularly useful way: the formalism will be constrained to *make physical properties manifest*. This is *precisely* the trait that will make such a symbol alphabet formalism useful, not just in MSYM but to advance the future of precision physics.

My project, then, will build off of this inspiration. From the wide array of possible symbol formalisms, this project will find one that is

1. Mathematically consistent,
2. Able to capture the full complexity of the functions, and
3. Able to capture the physical basis of MSYM’s surprising properties.

A.2

METHODOLOGY

A.2.1

WP1: MATHEMATICAL CONSISTENCY

The symbol alphabet proposed for tori [12] is inspired by a more general formalism developed by mathematicians. This formalism is referred to as **motivic**, with the idea that recurring mathematical structures are like musical *motifs* [19]. However, the existing alphabet does not fully follow this motivic formalism, and seems to contradict it in several respects: in particular by assigning a **weight** (a number which counts the complexity of the functions) different from that which would follow from the textbook definition in algebraic geometry [20]. The important question is whether this leads to a contradiction in the torus symbol alphabet formalism, or whether that formalism is consistent and just prioritizes different goals from the formalism proposed by mathematicians.

My team will begin this work package by finding the precise relation between the torus symbol formalism and the motivic formalism, in order to determine whether it is consistent and what it captures. Then, building off of

my results in ref. [13], we will characterize the space of all possible mathematically consistent symbol alphabet formalisms, enumerating the choices one can make in setting them up. Finally, we will use this characterization to find new alphabet formalisms with useful properties, informed by the other research of this proposal.

A.2.2

WP2: CAPTURING COMPLEXITY

One property that, as physicists, we would like weight to have is that it should increase for more complicated computations. This property forms a key organizing principle behind the current state of the art in differential equation methods for Feynman diagrams [5], allowing one to use these differential equations to systematically write down iterated integrals of rational functions.

A.2.3

WP3: LEARNING FROM MSYM

In order to capture the physical basis of the impressive structure found in alphabets in MSYM theory, my team will first characterize what that physical basis is. Two recent clues come from the structure of polynomials of kinematic invariants [21] and work building off of an old theorem of Lev Landau [22].

The first clue uses a mathematical formalism for polynomials in variables that obey relations. Here, the variables are kinematic variables describing particle scattering, and the relations are basic physical requirements such as momentum conservation. Ref [21] found that, under certain assumptions, a “fan” structure derived from these polynomials encodes the cluster algebra of alphabets in this theory. My group’s aim will be to relax these assumptions and see what continues to hold. As an example, we will check whether adding constants to the basis (in a way that obeys all required relations) changes the outcome. This work will involve using algebraic geometry software, likely Macaulay2, to construct “fan” structures with different assumptions.

Based on this investigation, it should be possible to pin down a minimal set of physically motivated requirements that give rise to the theory’s cluster algebra structure. I then plan for us to find a more direct derivation, building on an old theorem by Lev Landau about the singularities of scattering amplitudes [23], and understand how these minimal conditions arise from the rigorous derivation. As the picture solidifies, we will be able to characterize how it can apply to iterated integrals beyond rational functions, bringing in results from the other research directions of the project to obtain a consistent and powerful whole.

A.3

TEAM

This project will fund two PhD candidates and three three-year postdocs. The **first PhD candidate** will begin in the first year of the project. Their initial task will be to investigate fan structures obtained from different physical assumptions, using Macaulay2. They will then continue with the investigations of the third work package, while being exposed to, and having opportunities to get involved with, the other threads of inquiry. Based on my experience supervising advanced Master’s students, I expect this to be an ideal start for a PhD candidate: one that begins with a concrete task involving mathematical software, and uses it to build experience and intuition for bigger mathematical and physical questions. The student will benefit from mentorship not only from myself, but from the project’s postdocs, and as such I expect them to pick up quite valuable expertise for a strong start to a scientific career.

The **first postdoc**, hired in the first year of the project, will bring expertise with the motivic formalism. This will complement my expertise with the way physicists use symbol formalism and our priorities in doing so. Our goal will be to understand whether there is a risk of contradiction in the formalism physicists have applied to symbol alphabets beyond polylogarithms and to map the space of consistent alphabets, leading eventually to new useful formalisms as the work progresses in line with the first work package. They begin alongside the first PhD student, and I expect them to be a strong resource for that student’s development, able to explain concepts from algebraic geometry that the student is likely to encounter in their work.

The **second postdoc** will begin in the second year of the project, and will have expertise relevant to the second work package.

The **third postdoc** will join in the third year of the project. They will bring expertise in doing phenomenological calculations at scale in collider physics, gravitational waves, or both. They will be involved in each of the three work packages, and their role will be to see that the different threads of the project mature into useful tools for precision physics. This will involve generalizing insights gained from MSYM to Quantum Chromodynamics, finding symbol alphabet formalisms with properties directly useful to carrying out precision physics at scale, and refining the differential equations found earlier in the project for wider use. As the project comes to a close, they will be responsible for prototype software packages making the new techniques developed available to the wider community.

The **second PhD candidate** will join in the third year of the project, alongside the third postdoc. Like the third postdoc, they will be concerned with larger-scale computer code. They will initially focus on this in the context of the third work package, extending the first PhD student’s work to more challenging scattering amplitudes with more external particles, potentially bootstrapping amplitudes at higher loops. They will also collaborate with the third postdoc on building computer code implementing our new symbol alphabet

formalisms. Like the first PhD student, I expect that this early focus on computer code with the potential to expand conceptually later will be a strong start for a student, and they will be embedded in a lively group with many opportunities to learn.

A.4 RISK ANALYSIS

This project’s goal, of finding a formalism useful across many areas of precision physics, is ambitious, and I am well prepared to pursue it. I have experience with both substantial computational projects [24] and more mathematical investigations [13] and expertise on both the structure of amplitudes in MSYM and on iterated integrals beyond rational functions. I expect my experience supervising Master’s students and collaborating with junior scientists to transfer well to leading a scientific group, and I have chosen the team to complement me scientifically, adding expertise in particular areas of import (such as rigorous mathematics, Calabi-Yaus, and larger-scale phenomenology calculations).

For the first work package, there are two possible results to the initial investigation: either to show a flaw in the existing torus alphabet formalism, or to frame it in a clearer mathematical context that makes its consistency clear. The former would be quite important to point out and fix, as this is a formalism already in active use. The latter would enable more interaction between mathematicians and physicists on this topic, yielding more results of general use. Either will result in concrete criteria for generalizing the formalism.

For the second work package, the core foundation is present in the literature, requiring only my proposal’s novel approach to bring it together.

For the third work package, there is a risk that the observed connection between a “fan” structure of polynomials and cluster algebras is in some way a mathematical artifact. If so, the structure of this work package will, by attempting to generalize and ground the principles behind the observation, reveal this relatively quickly. Having thus learned something concrete about the implications of physical conditions on symbol alphabets in MSYM, it will then be straightforward to pivot, either to analyzing other promising recent approaches to this problem (for example see ref. [25]), or by linking to more rigorously confirmed conditions such as Landau’s theorem. In each case, this work package will provide useful input towards the overall goal.

A.5 IMPACT

The symbol alphabets developed in this project will have a direct impact on calculations in collider and gravitational wave physics. By building a reliable, general, and physically-motivated formalism, my team will significantly improve calculations of Feynman diagrams beyond rational functions, especially for cases beyond tori where no general formalism exists. The alphabet formalism I develop will lead, much as the formalism for rational iterated integrals led, to more efficient derivation of differential equations, series expansions in different regions of kinematic space, better cancellation of divergences due to recognizing identities between functions, and faster numerics. It will also yield significant progress towards understanding the mysteries observed in MSYM, exploring their implications outside of the specific mathematical formalism where they have appeared so far. By linking this formalism to the mathematical literature, I will also enable more interaction between mathematicians and physicists on this topic, leading to further breakthroughs.

The project would have a dramatic impact on the development of myself and the team I hire. It would position us as experts in a new and valuable formalism, one that has uses, and thus collaboration opportunities, across several fields of physics both across the European Union and around the world. It would allow me to establish a research group at the Institut de Physique Théorique with a clear, internationally recognized mission and a track record of developing useful tools for the community. It will give the team’s postdocs experience mentoring and interacting with PhD students, which will be valuable in their future careers, and it will give the PhD students a taste of many different types of expertise, giving them flexibility in what they pursue in future.

In the long run, the techniques I propose to develop will be essential for the high-energy physics community to fully address the new era of precision physics. If we are to be confident that the Large Hadron Collider and any future higher-energy colliders are exploited to their utmost, we will need precision theoretical predictions, and these will directly require new mathematical concepts. If we are to use gravitational wave telescopes to probe the behavior of neutron stars or to test theories beyond Einstein gravity, we will likewise require new ideas. A project of this kind *essential* if we are to advance our understanding of these aspects of the natural world.

REFERENCES

- [1] R. Boughezal *et al.*, “Theory Techniques for Precision Physics – Snowmass 2021 TF06 Topical Group Report,” [arXiv:2209.10639](https://arxiv.org/abs/2209.10639) [hep-ph].
- [2] A. Buonanno, M. Khalil, D. O’Connell, R. Roiban, M. P. Solon, and M. Zeng, “Snowmass White Paper: Gravitational Waves and Scattering Amplitudes,” in *Snowmass 2021*. 4, 2022. [arXiv:2204.05194](https://arxiv.org/abs/2204.05194) [hep-th].

- [3] A. B. Goncharov, M. Spradlin, C. Vergu, and A. Volovich, “Classical Polylogarithms for Amplitudes and Wilson Loops,” *Phys. Rev. Lett.* **105** (2010) 151605, [arXiv:1006.5703 \[hep-th\]](#).
- [4] S. Caron-Huot, L. J. Dixon, J. M. Drummond, F. Dulat, J. Foster, O. Gürdoğan, M. von Hippel, A. J. McLeod, and G. Papathanasiou, “The Steinmann Cluster Bootstrap for $N = 4$ Super Yang-Mills Amplitudes,” *PoS CORFU2019* (2020) 003, [arXiv:2005.06735 \[hep-th\]](#).
- [5] J. M. Henn, “Multiloop Integrals in Dimensional Regularization Made Simple,” *Phys. Rev. Lett.* **110** (2013) no. 25, 251601, [arXiv:1304.1806 \[hep-th\]](#).
- [6] J. L. Bourjaily *et al.*, “Functions Beyond Multiple Polylogarithms for Precision Collider Physics,” in *Snowmass 2021*. 3, 2022. [arXiv:2203.07088 \[hep-ph\]](#).
- [7] Z. Bern, J. Parra-Martinez, R. Roiban, M. S. Ruf, C.-H. Shen, M. P. Solon, and M. Zeng, “Scattering Amplitudes and Conservative Binary Dynamics at $\mathcal{O}(G^4)$,” *Phys. Rev. Lett.* **126** (2021) no. 17, 171601, [arXiv:2101.07254 \[hep-th\]](#).
- [8] J. L. Bourjaily, A. J. McLeod, M. von Hippel, and M. Wilhelm, “A (Bounded) Bestiary of Feynman Integral Calabi-Yau Geometries,” *Phys. Rev. Lett.* **122** (2019) no. 3, 031601, [arXiv:1810.07689 \[hep-th\]](#).
- [9] P. Lairez and P. Vanhove, “Algorithms for minimal Picard–Fuchs operators of Feynman integrals,” *Lett. Math. Phys.* **113** (2023) no. 2, 37, [arXiv:2209.10962 \[hep-th\]](#).
- [10] C. F. Doran, A. Harder, E. Pichon-Pharabod, and P. Vanhove, “Motivic geometry of two-loop Feynman integrals,” [arXiv:2302.14840 \[math.AG\]](#).
- [11] P. Lairez, E. Pichon-Pharabod, and P. Vanhove, “Effective homology and periods of complex projective hypersurfaces,” [arXiv:2306.05263 \[math.AG\]](#).
- [12] J. Brödel, C. Duhr, F. Dulat, B. Penante, and L. Tancredi, “Elliptic Symbol Calculus: from Elliptic Polylogarithms to Iterated Integrals of Eisenstein Series,” *JHEP* **08** (2018) 014, [arXiv:1803.10256 \[hep-th\]](#).
- [13] A. Forum and M. von Hippel, “A Symbol and Coaction for Higher-Loop Sunrise Integrals,” [arXiv:2209.03922 \[hep-th\]](#).
- [14] J. Golden and M. Spradlin, “The Differential of All Two-Loop MHV Amplitudes in $\mathcal{N}=4$ Yang-Mills Theory,” *JHEP* **1309** (2013) 111, [arXiv:1306.1833 \[hep-th\]](#).
- [15] J. Drummond, J. Foster, and Ö. Gürdoğan, “Cluster Adjacency Properties of Scattering Amplitudes in $\mathcal{N} = 4$ Supersymmetric Yang-Mills Theory,” *Phys. Rev. Lett.* **120** (2018) no. 16, 161601, [arXiv:1710.10953 \[hep-th\]](#).
- [16] L. J. Dixon, O. Gürdoğan, A. J. McLeod, and M. Wilhelm, “Folding Amplitudes into Form Factors: An Antipodal Duality,” *Phys. Rev. Lett.* **128** (2022) no. 11, 111602, [arXiv:2112.06243 \[hep-th\]](#).
- [17] L. J. Dixon, O. Gürdoğan, Y.-T. Liu, A. J. McLeod, and M. Wilhelm, “Antipodal Self-Duality for a Four-Particle Form Factor,” *Phys. Rev. Lett.* **130** (2023) no. 11, 111601, [arXiv:2212.02410 \[hep-th\]](#).
- [18] J. L. Bourjaily, Y.-H. He, A. J. McLeod, M. von Hippel, and M. Wilhelm, “Traintracks Through Calabi-Yaus: Amplitudes Beyond Elliptic Polylogarithms,” *Phys. Rev. Lett.* **121** (2018) no. 7, 071603, [arXiv:1805.09326 \[hep-th\]](#).
- [19] F. Brown, “Notes on Motivic Periods,” [arXiv:1512.06410 \[math.NT\]](#).
- [20] C. A. M. Peters and J. H. M. Steenbrink, *Mixed Hodge Structures*, vol. 52. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics, 2010.
- [21] L. Bossinger, J. M. Drummond, and R. Glew, “Adjacency for scattering amplitudes from the Gröbner fan,” [arXiv:2212.08931 \[hep-th\]](#).
- [22] L. Lippstreu, M. Spradlin, A. Yellespur Srikant, and A. Volovich, “Landau Singularities of the 7-Point Ziggurat II,” [arXiv:2305.17069 \[hep-th\]](#).
- [23] L. Landau, “On Analytic Properties of Vertex Parts in Quantum Field Theory,” *Nuclear Physics* **13** (1959) no. 1, 181 – 192. <http://www.sciencedirect.com/science/article/pii/0029558259901543>.
- [24] S. Caron-Huot, L. J. Dixon, F. Dulat, M. von Hippel, A. J. McLeod, and G. Papathanasiou, “Six-Gluon Amplitudes in Planar $\mathcal{N}=4$ super-Yang-Mills Theory at Six and Seven Loops,” *JHEP* **08** (2019) 016, [arXiv:1903.10890 \[hep-th\]](#).

- [25] S. He, X. Jiang, J. Liu, and Q. Yang, “On symbology and differential equations of Feynman integrals from Schubert analysis,” [arXiv:2309.16441 \[hep-th\]](#).
- [26] L. J. Dixon, J. M. Drummond, and J. M. Henn, “Bootstrapping the Three-Loop Hexagon,” *JHEP* **1111** (2011) 023, [arXiv:1108.4461 \[hep-th\]](#).
- [27] O. Steinmann, “Über den Zusammenhang zwischen den Wightmanfunktionen und der retardierten Kommutatoren,” *Helv. Physica Acta* **33** (1960) 257.
- [28] O. Steinmann, “Wightman-Funktionen und retardierten Kommutatoren. II,” *Helv. Physica Acta* **33** (1960) 347.
- [29] K. E. Cahill and H. P. Stapp, “OPTICAL THEOREMS AND STEINMANN RELATIONS,” *Annals Phys.* **90** (1975) 438.
- [30] J. L. Bourjaily, H. Hannesdottir, A. J. McLeod, M. D. Schwartz, and C. Vergu, “Sequential Discontinuities of Feynman Integrals and the Monodromy Group,” *JHEP* **01** (2021) 205, [arXiv:2007.13747 \[hep-th\]](#).
- [31] S. Caron-Huot and K. J. Larsen, “Uniqueness of Two-Loop Master Contours,” *JHEP* **1210** (2012) 026, [arXiv:1205.0801 \[hep-ph\]](#).
- [32] J. L. Bourjaily, E. Herrmann, and J. Trnka, “Prescriptive Unitarity,” *JHEP* **06** (2017) 059, [arXiv:1704.05460 \[hep-th\]](#).
- [33] W. D. Goldberger and I. Z. Rothstein, “An Effective field theory of gravity for extended objects,” *Phys. Rev. D* **73** (2006) 104029, [arXiv:hep-th/0409156](#).
- [34] M. Levi, A. J. McLeod, and M. Von Hippel, “NNNLO gravitational quadratic-in-spin interactions at the quartic order in G,” [arXiv:2003.07890 \[hep-th\]](#).
- [35] M. Levi and J. Steinhoff, “Spinning gravitating objects in the effective field theory in the post-Newtonian scheme,” *JHEP* **09** (2015) 219, [arXiv:1501.04956 \[gr-qc\]](#).
- [36] S. Caron-Huot, L. J. Dixon, M. von Hippel, A. J. McLeod, and G. Papathanasiou, “The Double Pentaladder Integral to All Orders,” *JHEP* **07** (2018) 170, [arXiv:1806.01361 \[hep-th\]](#).

B CURRICULUM VITAE AND TRACK RECORD

B.1 PERSONAL DETAILS

Family name, First name: von Hippel, Matthew

Researcher unique identifier (ORCID): 0000-0002-6328-2936

URL for web site: <https://www.ipht.fr/Phocsa/Membres/Annuaire/index.php?uid=mvonhippel>

EDUCATION

- 23/05/2014 PhD: “Explorations in Planar $\mathcal{N} = 4$ super Yang-Mills”
Department of Physics and Astronomy, Stony Brook University, USA
Advisor: Michael Douglas
- 2010 Bachelor of Science - Summa Cum Laude
Department of Physics, Tufts University, USA

CURRENT POSITION

- 2023– Permanent Researcher
Institute for Theoretical Physics, CEA Paris-Saclay, France

PREVIOUS POSITIONS

- 2021–2023 Assistant Professor
Niels Bohr International Academy, University of Copenhagen, Denmark
- 2017–2021 Postdoctoral Fellow
Niels Bohr International Academy, University of Copenhagen, Denmark
- 2014–2017 Postdoctoral Fellow
Perimeter Institute, Canada

B.2 RESEARCH ACHIEVEMENTS AND PEER RECOGNITION

RESEARCH ACHIEVEMENTS

I have authored **26 research papers** (22 published) and 4 conference proceedings, with over **1400 citations**¹. Below I list the ten I view as most significant, with a brief discussion of the context for each. As is standard in my field, authors are listed alphabetically. None of the below papers were written with my PhD supervisor.

- Lance J. Dixon, James M. Drummond, **Matt von Hippel**, and Jeffrey Pennington, “Hexagon functions and the three-loop remainder function”, *JHEP* 1312, 049 (2013), arXiv:1308.2276 [hep-th] (cited 136 times)

During my PhD I visited the SLAC National Accelerator Laboratory for several months, collaborating with Lance Dixon and his group. At the time, the three-loop six-particle amplitude in MSYM was known at the level of the symbol, but not fully, lacking terms proportional to transcendental constants [26]. It was my task to build an implementation of the relevant space of functions that could be used to fix these missing terms, expressing the final answer for the amplitude in a form suitable for numerical evaluation. This work initiated the “hexagon function program”, which in subsequent years would see me compute six-particle amplitudes in this theory up to seven loops.

- Simon Caron-Huot, Lance J. Dixon, Andrew McLeod, and **Matt von Hippel**, “Bootstrapping a Five-Loop Amplitude Using Steinmann Relations”, *Phys.Rev.Lett.* 117 (2016) no.24, 241601, arXiv:1609.00669 [hep-th] (cited 131 times)

A milestone in the hexagon function program, in this work I employed a result from the history of quantum field theory called the Steinmann relations [27, 28, 29] as additional constraints on the amplitude, allowing the attainment of five-loop precision. This work reintroduced the community to the Steinmann relations, leading to a renaissance of work using and investigating them (e.g., ref. [30]).

¹Data retrieved as of September 4, 2023 from the inSPIRE database. My unique author identifier there is M.Von.Hippel.1

- Jacob L. Bourjaily, Andrew J. McLeod, Marcus Spradlin, **Matt von Hippel**, and Matthias Wilhelm, “The Elliptic Double-Box Integral: Massless Amplitudes Beyond Polylogarithms”, *Phys.Rev.Lett.* 120 (2018) 121603, arXiv:1712.02785 [hep-th] (cited 88 times)

Based on the earlier observation that a cut of a particular integral, called the double-box, in MSYM involved a torus geometry [31], it was expected to be possible to express the full integral in terms of iterated integrals of rational functions with one algebraic piece: a factor dividing by the square root of a polynomial defining the geometry. During a research stay at the Institute for Advanced Study I computed the relevant integrals. Interestingly this result does not naturally make manifest the symmetries of the diagram, which sharpened the need for new representations of Feynman integrals involving tori.

- Jacob L. Bourjaily, Yang-Hui He, Andrew J. McLeod, **Matt von Hippel**, and Matthias Wilhelm, “Train-tracks Through Calabi-Yaus: Amplitudes Beyond Elliptic Polylogarithms”, *Phys.Rev.Lett.* 121 (2018) no.7, 071603, arXiv:1805.09326 [hep-th], (cited 81 times)

This paper was the beginning of my involvement with Feynman diagrams that involve Calabi-Yaus. Along with my group at the Niels Bohr Institute, we demonstrated that a family of diagrams appearing in MSYM involves Calabi-Yaus of arbitrarily high dimension. I was in particular responsible for checking that these Calabi-Yaus are in fact irreducible, and thus the integrals cannot be represented by simpler functions, to high orders. For certain sets of external particles, each of these diagrams is the only contribution to the corresponding amplitude, meaning that they cannot be canceled by other contributions. This conclusively established that a full understanding of MSYM would require understanding of Calabi-Yaus, inspiring much of the research leading to this proposal.

- Jacob L. Bourjaily, Andrew J. McLeod, **Matt von Hippel**, and Matthias Wilhelm, “A (Bounded) Bestiary of Feynman Integral Calabi-Yau Geometries”, *Phys.Rev.Lett.* 1122 (2019) 031601, arXiv:1810.07689 [hep-th] (cited 46 times)

Working with a class of integrals Jacob Bourjaily was investigating in connection with a different research programme [32], I was able to prove that a particular class of integrals, termed “marginal”, were intimately linked to Calabi-Yaus. For each integral in this class, the first possible obstruction to integration could be shown to define a Calabi-Yau. I found all-loop families of integrals that displayed this behavior. The simplest such integral, called the two-loop “tardigrade” for its resemblance to the hardy microscopic organism of that name, involves a two-dimensional Calabi-Yau (a K3), making it the most geometrically complex two-loop integral discovered to date. This integral can represent scattering of two massless particles to form three massive ones, and as such will be relevant to precision measurements of the quadratic gauge coupling and the top quark Yukawa coupling at the Large Hadron Collider and future colliders, sharpening the need for an understanding of these kinds of integrals.

- Simon Caron-Huot, Lance J. Dixon, Falko Dulat, **Matt von Hippel**, Andrew J. McLeod, Georgios Papathanasiou, “The Cosmic Galois Group and Extended Steinmann Relations for Planar N=4 SYM Amplitudes”, *JHEP* 1909 (2019) 061, arXiv:1906.07116 [hep-th], (cited 88 times)

In this paper and its companion, the hexagon function program reached seven-loop amplitudes. This was made possible by two observations. One, referred to as the “cosmic Galois group”, is an iterative structure that constrains amplitudes to be built out of functions that occur in lower-loop amplitudes. The other, referred to here as Extended Steinmann relations, are a relationship between the alphabet letters appearing in different entries in the symbol, forming a kind of generalization of the Steinmann relations used previously. These extended relations were linked by others to cluster algebras, a link which has become both an extremely fruitful area of research and a major source of motivation for this proposal.

- Michèle Levi, Andrew J. McLeod, **Matthew von Hippel**, “NNNLO gravitational spin-orbit coupling at the quartic order in G”, arXiv:2003.02827 [hep-th] (cited 26 times)

The recent success of LIGO has seen a renaissance in methods to compute gravitational waveforms. As two astrophysical objects approach each other before a merger, their behavior is well-described by a type of quantum field theory approach called effective field theory [33]. In this paper and its companion [34] I pushed beyond the state of the art for the dynamics of spinning gravitational binaries, using the effective field theory for gravitating spinning objects [35] to compute all contributions at the quartic order in Newton’s constant, corresponding to three-loop diagrams. This work gave me experience with phenomenological computations at scale, which has informed my research priorities since, as well as serving as my entry into the gravitational-wave community, making me increasingly aware of the precision demands of that research program.

- Hjalte Frellesvig, Cristian Vergu, Matthias Volk, **Matt von Hippel**, “Cuts and Isogenies”, *JHEP* 05 (2021) 064, arXiv:2102.02769 [hep-th], (cited 9 times)

In this paper, I was inspired to investigate what seemed to be a weakness of the existing formalism for Feynman diagrams involving torus geometries. In the current formalism, iterated integrals on a torus are

defined in terms of a single specific geometry, and this does not straightforwardly allow comparisons to integrals defined with other geometries, even other different tori. However, one of the most well-studied Feynman diagrams of this kind, the two-loop sunrise, had two presentations in the literature corresponding to two different torus geometries. This work clarified the relation between these two approaches, showing that similar phenomena can occur in other diagrams. We argued that only one of these tori is fundamental to the diagram, however the other can be useful in many situations, which has inspired me to speculate about future symbol formalisms that could encompass both geometries in a single framework.

- Andreas Forum, **Matt von Hippel**, “A Symbol and Coaction for Higher-Loop Sunrise Integrals”, *SciPost Phys. Core* 6, 050 (2023), arXiv:2209.03922 [hep-th], (cited 9 times)

At the Niels Bohr Institute, I prioritized mentoring students. In practice, this took the form of supervising Master’s theses, and over the course of three years I supervised three such students, each thesis resulting in a paper. The last of these covered particularly interesting ground. Pursuing the same methodology used to derive the torus symbol formalism, we found that a wide variety of other formalisms could be derived for diagrams involving Calabi-Yaus. This profusion of options is a key motivation for this proposal, and a central task of the proposal will be building on this work, systematically characterizing each choice made in the formalism and describing the full space of mathematically consistent symbols that can be devised.

- Andrew McLeod, Roger Morales, **Matt von Hippel**, Matthias Wilhelm, Chi Zhang, “An Infinite Family of Elliptic Ladder Integrals”, *JHEP* 05 (2023) 236, arXiv:2301.07965 [hep-th], (cited 3 times)

This work describes a class of Feynman diagrams that appear to all orders but always contain the same torus. I characterized the differential equations these functions satisfy, with the motivation of understanding them in a similar vein to another infinite family of integrals that I and collaborators characterized to all orders [36]. These diagrams should be valuable because, due to their differential equation structure, they are likely to have a consistent symbol alphabet to higher orders. A key test of the formalism I develop in this proposal will be how it represents these integrals, in particular whether it can describe them in a simple way amenable to all-orders generalizations.

PEER RECOGNITION

From August 2018 to August 2020, I was supported by a **Marie Curie Individual Fellowship** from the European Union, with the aim of probing the limitations of bootstrap methods for perturbative amplitudes. Under this grant I pursued integrals that stretched the boundaries of what could be bootstrapped at the time. In particular, that work led me to much greater involvement with iterated integrals beyond rational functions, and thus to the motivation behind this proposal.

I am regularly invited to give seminars, colloquia, and conference talks. Below I list a selection of invited talks I have given at international conferences:

- **Elliptic Integrals in Fundamental Physics**, Mainz Institute for Theoretical Physics, September 2022: “Strange Symbol Structures for a Sufficiently Supplemented Sunrise”
- **Antidifferentiation and the Calculation of Feynman Amplitudes**, DESY Zeuthen, October 2020: “Direct Integration for Multi-leg Amplitudes: Tips, Tricks, and When They Fail”
- **Scattering Amplitudes and the Conformal Bootstrap**, Aspen Center for Physics, August 2019: “Ideas in Amplitudes for Non-Amplitudes People”
- **Amplitudes 2019**, Trinity College Dublin, July 2019: “Hexagon Functions to Seven Loops”
- **Nonperturbative Methods for Conformal Theories**, Natal, April 2019: “The Double Pentaladder Integral for Finite Coupling”
- **Amplitudes in the LHC Era**, Galileo Galilei Institute, November 2018, “Calabi-Yaus and Other Animals”
- **Amplitudes 2015**, Zurich, July 2015, “Hexagon Functions: Bootstrapping an NMHV Amplitude through Four Loops”

Other highlights include a colloquium at Oregon State University in November 2015, relatively early in my career, and a guest lecture in a quantum field theory course organized among mathematicians at University of California Berkeley in August 2020.

B.3 ADDITIONAL INFORMATION

OTHER CONTRIBUTIONS TO THE RESEARCH COMMUNITY

I have organized several conferences for my subfield, both online during the COVID era and later in person. I have raised the public profile of both my subfield and the wider area of high-energy physics via my outreach activities. These most principally include my blog at 4gravitons.com, which typically receives one hundred unique visitors per day and consists of weekly posts where I explain science and the scientific process to the general public. This blog has put me in frequent contact with journalists, who regularly contact me with questions about my subfield. Occasionally I write pieces in journalistic outlets myself, to draw public attention to the valuable work done by my research community and encourage public support. I have written for Ars Technica, Scientific American Magazine, and Quanta Magazine.